CLASS 9 — SCIENCE CHAPTER TEST SET A

Chapter 11: **Patterns in Life: Diversity and Classification** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The variety of living organisms on the Earth is called: 1

(a) classification **(b)** biodiversity **(c)** ecosystem **(d)** evolution

2. Aristotle grouped animals mainly on the basis of: 1

(a) cell wall **(b)** DNA
(c) habitat — land, water and air **(d)** number of kingdoms

3. Cyanobacteria are: 1

(a) autotrophic prokaryotes **(b)** fungi
(c) heterotrophic eukaryotes **(d)** viruses

4. Fungi obtain food by: 1

(a) absorbing nutrients from dead organic matter
(b) photosynthesis
(c) ingestion
(d) making food using chlorophyll

5. Spirogyra is an example of: 1

(a) pteridophyta **(b)** bryophyta
(c) angiosperm **(d)** algae (thallophyta)

6. Pine and cycas are: 1

(a) pteridophytes **(b)** angiosperms **(c)** gymnosperms **(d)** bryophytes

7. Animals without a notochord are called: 1

(a) non-chordates **(b)** vertebrates **(c)** protochordates **(d)** chordates

8. Earthworm belongs to the phylum: 1

(a) Mollusca **(b)** Nematoda **(c)** Annelida **(d)** Arthropoda

9. The group of vertebrates that is cold-blooded and breathes through gills is: 1

(a) birds **(b)** fishes **(c)** reptiles **(d)** mammals

10. The two-part naming system was introduced by: 1

(a) Carolus Linnaeus **(b)** Aristotle **(c)** Whittaker **(d)** Darwin

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. What is biodiversity? Give one reason for its importance. 2

12. What is classification? Why is it needed? 2

13. Name any four characteristics used for classifying organisms. 2

14. Name the five kingdoms of Whittaker. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Explain why biodiversity is important with two examples. 3

16. What are biodiversity hotspots? Name two in India. 3

17. State three advantages of classification. 3

OR

How did the classification systems develop from Aristotle to Whittaker? 3

18. State the basis of the five kingdom classification. 3

OR

Describe kingdom Monera with two examples and uses. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) What is biodiversity? 5
(ii) Why should it be conserved?
(iii) Give two examples of interdependence among organisms.

OR

Case study: Pakke Tiger Reserve has nearly 300 bird species and four hornbill species that use different fruits. 5
(i) What does it show about diversity?
(ii) How do hornbills avoid competition?
(iii) Why are hotspots important?
(iv) Suggest two conservation measures.

20. (i) Why do we classify organisms? 5
(ii) State the criteria used.
(iii) Describe the five kingdom system in brief.

OR

Case study: A student examines a drop of pond water and sees Euglena, Paramecium and Spirogyra. 5
(i) Place each in its kingdom/group.
(ii) Which can make food?
(iii) Which moves with cilia?
(iv) Which body type is a thallus?

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND